PIN OUT

J1 Conn_02x30_Row_Letter_First

J2 Conn_02x30_Row_Letter_First

H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
H5 MountingHole
H6 MountingHole
H7 MountingHole
H8 MountingHole

MICROCONTROLLER

U3 XIAO-SAMD21-SMD

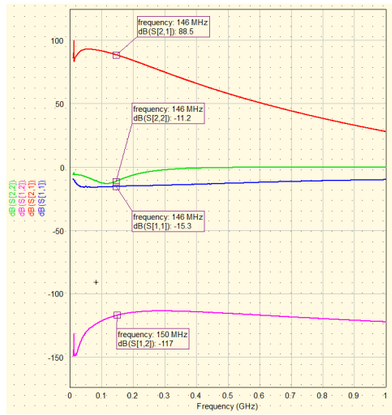
PA02_A0_D0 · PA4_A1_D1 · PA10_A2_D2 · PA11_A3_D3 · PA8_A4_D4_SDA · PA9_A5_D5_SCL · PB08_A6_D6_TX
PA31_SWDIO · PA30_SWCLK · RESET · GND · VBUS · GND · 3V3 · PA6_A10_D10_MOSI · PA5_A9_D9_MISO · PA7_A8_D8_SCK · PB09_A7_D7_RX · VIN · GND

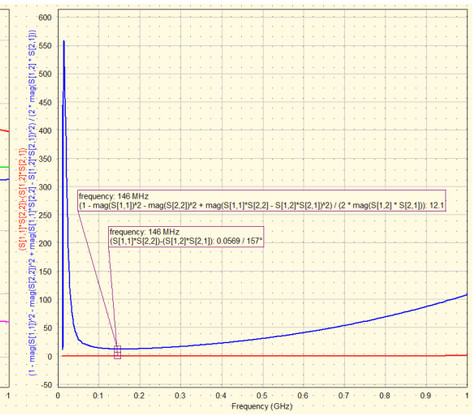
D2 LED · D3 LED · D4 LED
R4 1k · R5 1k · R12 1k
R2 49.9 · R1 49.9 · R3 49.9
DATA · LNA_ENABLE · CLK · LE · PLL_ENABLE

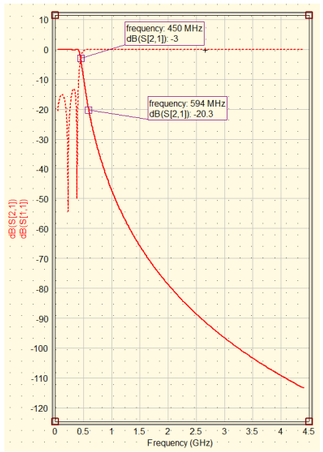
IC1 ADP7118AUJZ-5.0-R7
VIN · GND · EN · SENSE · VOUT
POWER_IN · C8 2.2uF · C24 2.2uF · 5V

D5 SS14

Capacitor, 2.2µF, 0603, 10 V, X5R







PCB Stackup & Fabrication ParametersLayer Count:
4Board Thickness: 1.6 mm
Base Material: FR4Layer Order: Top (RF/Signal) | In1 (GND) | In2 (VCC) | Bottom (Signal)
Outer Copper Weight: 1 oz (0.035 mm)
Inner Copper Weight: 0.5 oz (0.015 mm)
RF Dielectric (Top to In1): 0.2 mm (Prepreg 7628, Er: approx 4.4)RF
JLCPCB Profile: JLC04161H-7628 (Standard)
Impedance Control Service: No
NoSurface Finish: ENIG

LOW NOISE AMPLIFIER

IC2 ADP7118AUJZ-5.0-R7
POWER_IN · C2 2.2uF · C4 2.2uF · C32 1.5nF · 5V_LNA · R17 7.5k · R18 10k · LNA_ENABLE
Capacitor, 2.2µF, 0603, 10 V, X5R

U2 TCCH-80+
DC · NOT_USED_1 · NOT_USED_2 · RF-IN_&_DC
L1 180nH · C33 10pF · R16 49.9 · C30 1nF

IC6 PSA-5454+
GND_1 · GND_2 · RF-IN · RF-OUT · GND_3 · BIAS
J4 · C31 1nF · GND Conn_Coaxial_Small

R27 10 · C52 10nF · C49 100pF · L2 180nH · C37 1nF · R10 3.9k · R22 150 · C39 1nF · R19 1.5k · R20 10 · C38 1nF · R21 10 · Q1 2SC3356

R32 10 · C51 10nF · C50 100pF · L9 180nH · C43 1nF · R23 3.9k · R24 150 · C44 1nF · R25 1.5k · R26 10 · C45 1nF · Q2 2SC3356

L10 180nH · C46 1nF · R28 3.9k · R29 150 · C47 1nF · R30 1.5k · R31 10 · C48 1nF · Q3 2SC3356

L8 56nH · C40 5.6pF · C41 5.6pF

Bias 2SC3356:
NF=1dB
VCE=1V
ICQ=25mA

FILTER

L6 27nH · L7 27nH · C42 180pF · C27 10pF · C28 16pF · C29 10pF · J3 · Conn_Coaxial_Small

Low-pass filter, 440MHz cutoff
5th order
Chebyshev , PI-type,
Impedance 50

Filter Components:
Dielectric NPO/C0G
Case 0402
Tolerance Cap 1% / Ind 2%
SRF 3.5GHz
GJM1555C1H100FB01
GJM1555C1H160FB01
LQW15AN27NG00

LOCAL OSCILATOR

IC5 ADP150AUJZ-3.3-R7
5V · C9 1uF · C1 1uF · 3V3P
Capacitor, 1 µF, 0603, 10 V, X5R

C5 0.1uF · C15 0.1uF · C7 10pF · DVDD
C6 0.1uF · C16 0.1uF · C12 10pF · SDVDD · VP
C10 0.1uF · C34 10pF · VREF
C11 0.1uF · C26 10pF · VTEMP
C35 0.1uF · C36 10pF · VCOM
Decoupling Capacitors:
Ceramic capacitor, 0.1 µF, 0402, 16 V, X7R
Ceramic capacitor, 10 pF, 0402, 50 V, NPO

C18 0.1uF · C17 10pF · VVCO

U1 ADF4351
VP · AVDD · VVCO · DVDD · SDVDD · LD · MUXOUT · RF_OUT_A+ · RF_OUT_A- · RF_OUT_B+ · RF_OUT_B- · VTUNE · CPout · SW · CLK · DATA · LE · CE · PDB_RF · REFIN · TEMP · VCOM · VREF · RSET · CPGND · AGND · AGNDVCO · DGND · SDGND · EP

CLK · DATA · LE · PLL_ENABLE · R6 10k · R7 10k · R8 10k · R9 10k

Y1 ECS-TXO-2520MV-100-AY-TR
VDD · TRI-STATE · OUTPUT · GROUND · C3 1nF · C25 100pF · C19 1uF
Ceramic capacitor, 1 nF, 0603, 50 V, NPO

R14 5.1k · D1 LED · R11 1k

L3 120nH · L4 30nH · L5 30nH · C13 5.6pF · C14 120pF · C20 5.6pF · C21 1.1nF · R15 464 · C23 33nF · R13 2.94k · C22 180pF

LC blaun:
Dielectric NPO/C0G
Case 0402
Tolerance 2%
SRF 3.5GHz
Q >47

C 5.6pF GJM1555C1H5R6WB01D
L 30nH LQW15AN30NG00D
LDC 120nH LQW15AWR12G80D
CDC 120pF GRM1555C1H121JA01D

Filter Components:
Dielectric NPO/C0G
Case 0603 (C2 is 0805)
Tolerance Res 1% / Cap 5%
RC0603FR-07464RL
RC0603FR-072K94L
GCM1885C1H112JA16D
GCM21B5C1H333JA16L
GCM1885C1H181JA16D

MIXER

IC3 ADE-5+
GROUND_1 · IF · RF · LO · GROUND_3 · GROUND_2

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FCEFyN FACULTAD DE CIENCIAS EXACTAS, FÍSICAS y NATURALES

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Title: TRANSPONDER CUBESAT VHF/UHF

Size: A3	Date: 2024-11-25	Rev: A
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